DIP-CSE340 ASSIGNMENT 4
Deadline : 22/11/2024

Total Marks : 40 Marks **Total Days : 15 Days**

Instructions:

- Standard institute plagiarism policy holds.
- Please do submit your assignments at least before 11:50 PM on the deadline date to avoid any unexpected issues.
- The Kaggle competition will close after the deadline in the classroom
- Submit files : *report_Rollnumber.pdf* and Jupiter notebook *code_Roll_Number.ipny*.
- Your code and report with the images generated will be checked before you are graded.
- Uploading the original images will result in an error. So if you can achieve a 0 score, mail regarding the same.

Section A

20 marks

The compression of an image (grayscale) has two parts: An encoder and a decoder. This Kaggle assignment will evaluate you based on the compressed and reconstructed images. See the Figure 2 for reference of the pipeline. You can design your own encoder/decoder pipeline and note that the output of the encoder should be a bit stream and not floating-point/integer numbers.

1. You are provided with two RBG.tiff images $1200 \times 1200 \times 3$ with the watermark figure 1(b).
   (a) Remove the watermark from the image (by coding and not by tools like Adobe)
   (b) Convert the images to Grayscale
   (c) Compress the Grayscale images using 'Encoder' designed in Q2.
   (d) Decompress the Grayscale image to size (1200×1200) using the 'Decoder' designed in Q2.

   Note: Even if you cannot remove the watermark. You can proceed with the encoder and decoder. The original image without the watermark is not provided to you, and the same is stored as a .tiff image. The evaluation is done using its grayscale image of the tiff image.

(a) Image (b) Logo

Figure 1: (a) Example image and (b) Watermark image

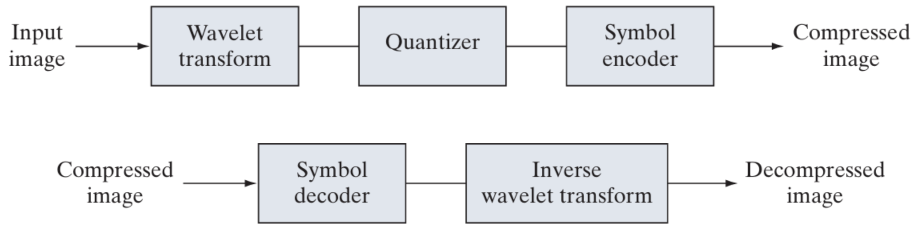

Figure 2: The image compression pipeline as in the DIP textbook page 615

2. Write two functions/classes/modules (Encoder and Decoder) that take an image and perform the respective operations to give the output. The function should be agnostic to similar images (i.e., you cannot use it if-and-else perform different operations on the images (Image1 and Image2)). (You can have an additional function, such as a function to remove the watermark and a main function that calls the other functions.) Upload the code in the classroom as a Jupiter notebook *code_Roll_Number.ipny*. A missing code file will result in zero marks in the assignment.

   1. Convert .tiff image → Grayscale → numpy array of grayscale
   2. Input numpy array of grayscale → Encoder → Output bitstream in numpy array
   3. Input Bitstream in numpy array → Decoder → Output 1200 x 1200 Reconstructed image

3. Beat the compression score in the Kaggle challenge. You are fighting against each one in the class.

   (a) The images in figure 1(a) is Image1 and a similar Image2 is provided with watermark.

   (b) The metrics used for the evaluation are :

   - mean squared error (MSE) between the Reconstructed Image and the original image.

$$SUM = \sum_{i=1}^{2} MSE(Image_i, RestoredImage_i)$$

   - compression ratio (ρ) between the compressed image and the original image.

$$\rho = \sum_{i=1}^{2} \frac{\text{size of compressed } Image_i}{\text{size of } Image_i} = \sum_{i=1}^{2} \frac{\text{size of encoded stream in bits of } Image_i}{1200 \times 1200 \times 8 \text{ bits}}$$

   - combined metric (the smaller the values, the better the result):

$$score(Image1, Image2) = SUM + (200)^{\rho}$$

   Note : Score metric ranges from 1. The best value is 1 when $SUM = 0$ and as $\rho \to 0$.

   (c) You can use any library, inbuilt function, or code base but not deep learning.

   (d) Submissions allowed per day 30.

Note : It should be possible to input the bitstreams of the compressed image uploaded in the kaggle challenge into the decoder to recreate the reconstructed image.

4. A tie in the competition will result in those with better presentation and techniques getting higher marks. For the same, the code and its time complexity will also be considered.

Section B

20 marks

Submit a latex report

1. Explain the method used for erasing the watermark. 6 marks

2. Write the documentation and pseudo-code of the pipeline used for the image compression challenge (both the encoder and decoder).Do include details on how to use the bitstream uploaded in the kaggle challenge to reconstruct the image uploaded with it. 8 marks

3. Show the change in the output of the grayscale image at the different stages of the image compression (see the pipeline in figure 2). Plot the histogram of the original (given image after removing watermark) and decompressed grayscale images. 4 marks

4. Explain any additional method and/or pre/post-processing techniques you used to improve the quality of your output as you progressed in the challenge. 2 marks

END